VHDL Not Gate Simulation

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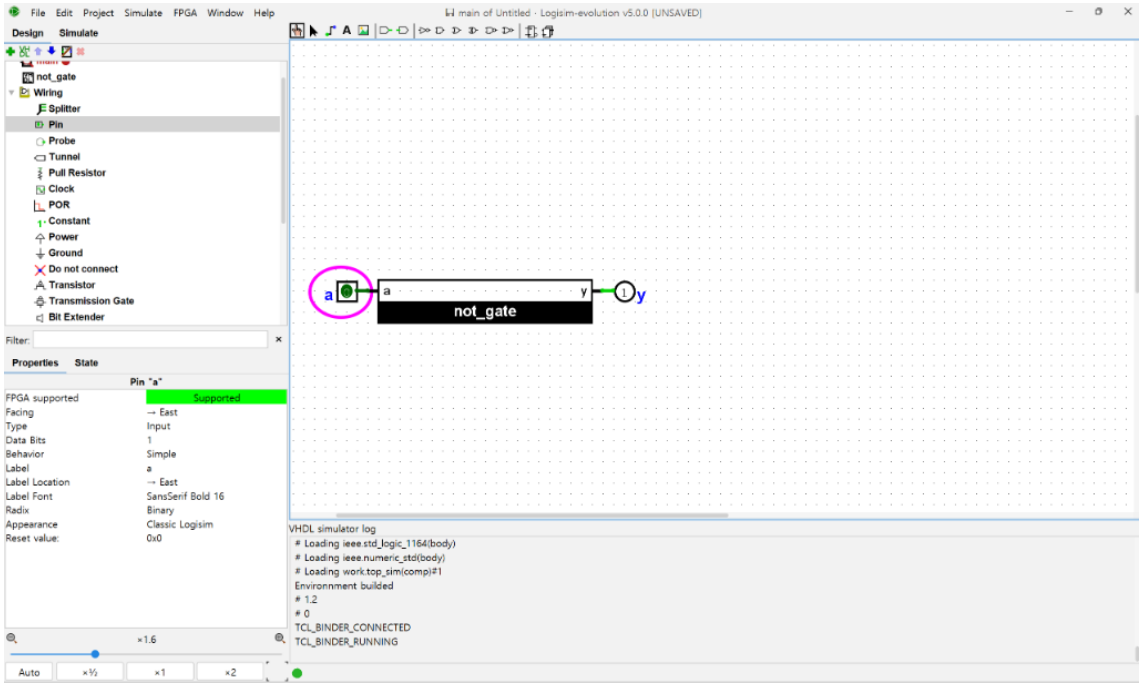
1. VHDL Code

```
LIBRARY ieee;
USE ieee.std_logic_1164.ALL;

ENTITY not_gate IS
    PORT (
        a : IN std_logic;
        y : OUT std_logic
    );
END not_gate;

ARCHITECTURE Behavioral OF not_gate IS
BEGIN
    y <= NOT a;
END Behavioral;
```

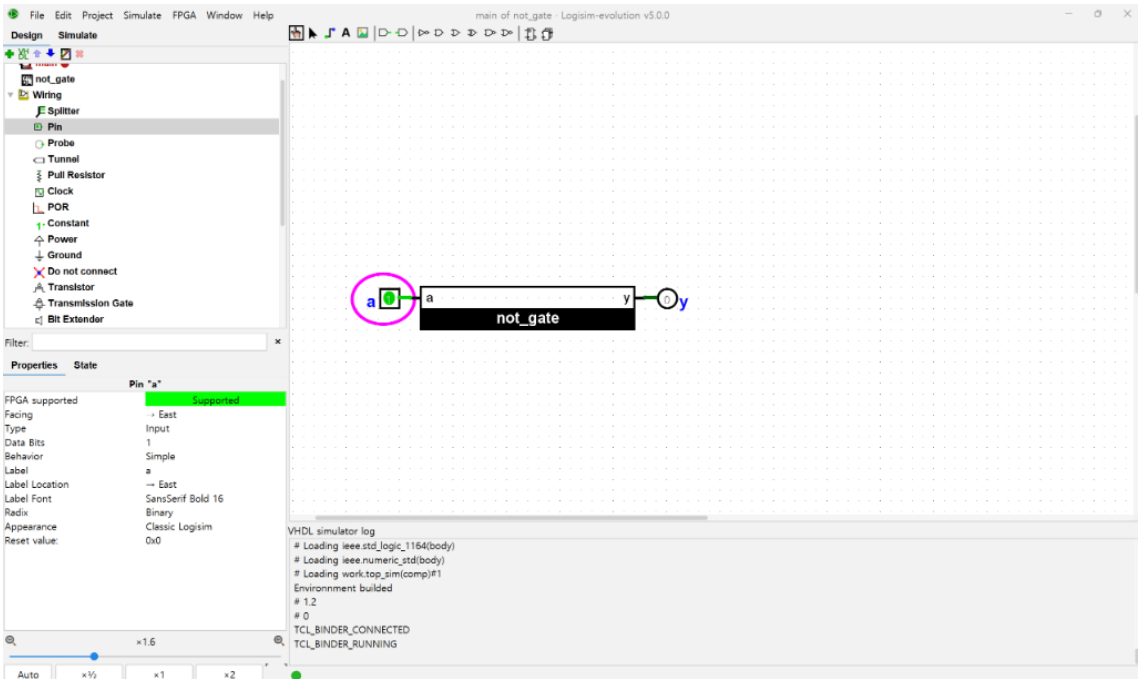
2. Logisim-evolution Simulation



Input : a = 0

Output : y =1

When the input a is 0, the output y becomes 1.



Input : a = 1

Output : y = 0

3. Simulation Results

When the input a is 1, the output y becomes 0.

Input a	Output y
0	1
1	0

4. Conclusion

The simulation results match the truth table of a Not gate. Therefore, the VHDL NOT gate was implemented and simulated successfully in Logisim-evolution using Questa.